Intelligent Ultrasonic Sensing for Movement Intensity Classification using Hybrid Machine Learning

Master of Engineering Information Technology
Module: Individual Project
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***Abstract*—Ultrasonic sensing provides a low-cost approach to detecting human presence and distinguishing human activity from non-human environmental conditions. The study evaluates how signal representation, model architecture, environment, position, and subject affect classification performance. Classical machine learning models and convolutional neural networks (CNNs) were evaluated using full and position normalized signal representations across two environments. For human presence detection, the results show that cross environment generalization is strongly influenced by the non human condition: ChairTable was substantially more challenging than EmptyRoom. The position normalized CNN1D model achieved 97.72% accuracy on ChairTable, compared with 55.14% for the best classical shape invariant model and 33.53% for the position normalized CNN2D model. For the Standing versus Reading task, CNN models achieved high performance, with CNN2D reaching 96.86% accuracy using position normalized signals and 97.13% using position aware signals. The results also show that normalization can reduce the influence of position and amplitude while changing performance on the Reading activity. Overall, the findings demonstrate that ultrasonic signals contain sufficient information for human presence and activity classification, while performance depends strongly on signal representation and environmental conditions. The study also highlights the importance of evaluating non human conditions separately and controlling for positional effects when interpreting ultrasonic sensing results.**

***Index Terms*—Ultrasonic Sensing; Human Presence Detection; Human Activity Recognition; Deep Learning; Convolutional Neural Networks; Classical Machine Learning; Signal Representation; Position Normalization; Cross Environment Generalization; Standing vs Reading Classification.**

I. Introduction

Ultrasonic sensing has potential applications in human-presence detection and activity recognition because reflected acoustic signals contain information about the interaction between a transmitted waveform and the surrounding environment. Traditional sensing approaches often rely on cameras or other wearable and environmental sensors. Ultrasonic sensing provides an alternative approach because it can capture information about objects and human movement through reflected acoustic signals without requiring a camera module representation.

In this Individual Project, ultrasonic sensor data is investigated for two related classification problems: detecting human presence and distinguishing between standing and reading activities. The study focuses not only on classification accuracy but also on how the choice of signal representation, machine learning model, environment, and subject affects performance. In particular, ultrasonic measurements can contain variations caused by distance and signal amplitude. These variations may provide useful information for classification, but they can also act as unwanted positional cues. Therefore, evaluating both position aware and position-normalized representations is important for determining whether a model is learning characteristics related to the target activity or relying on environmental and positional differences.

The first part of the project investigates human presence detection by distinguishing human conditions from non-human conditions, including an EmptyRoom condition and a ChairTable condition. The experiments compare classical machine-learning approaches with convolutional neural networks (CNNs), using different representations of the ultrasonic signals. Performance is evaluated not only on overall accuracy but also across individual experimental conditions and between environments. This makes it possible to examine whether a model that performs well during development also generalizes to a different environment.

The second part investigates whether ultrasonic signals can distinguish between a person who is Standing and a person who is Reading. This experiment is designed to examine classification after controlling for positional and amplitude related effects. Both position aware and position normalized representations are evaluated using classical models and CNN architectures. The results are also examined across individual subjects to determine whether performance is consistent between participants.

The project therefore addresses several aspects of ultrasonic sensing: human presence detection, comparison of classical machine learning and deep learning, the influence of signal representation, cross environment generalization, robustness to

different non-human conditions, and Standing versus Reading classification. By evaluating these factors separately, the study aims to provide a clearer understanding of what information in the ultrasonic signal contributes to classification performance and where the main limitations of the sensing approach occur.

II. Related work

Ultrasonic sensing has been widely investigated for detecting the presence, position, and movement of people in indoor environments. More recent approaches incorporate machine learning to improve detection performance by learning patterns from ultrasonic measurements rather than relying solely on fixed thresholds. Particular attention is given to signal representations, environmental generalization, and the use of deep learning models for extracting discriminative patterns from ultrasonic data.

A. Traditional Approaches to Human Detection Using Sensors

Human detection has traditionally been implemented using sensing modalities such as cameras, passive infrared (PIR) sensors, radar, and ultrasonic sensors. Vision module systems can provide detailed spatial information, but they can be sensitive to lighting conditions and raise privacy concerns in environments where continuous imaging is undesirable. Alternative sensing technologies can avoid some of these limitations by detecting physical characteristics of a person without directly acquiring visual information.

Ultrasonic sensing is particularly relevant because it measures reflected acoustic energy from objects within the sensing field. Prior systems have demonstrated this using supervised learning over segmented reflection patterns for still-presence sensing [1] and wide-band reflection superposition for ultrasonic occupancy counting [2]. Changes in the received echo can indicate the presence, position, movement, or activity of a target. However, the reflected signal is influenced not only by the human target but also by the surrounding environment, target geometry, distance, and surface properties. Consequently, a system that performs well under a controlled experimental configuration may not necessarily generalize to a different environment. This concern is echoed in recent sensor technology literature, where passive infrared, thermal, radar, and ultrasonic modalities are each shown to trade off accuracy, cost, and privacy differently depending on deployment conditions, with no single modality dominating across all occupancy monitoring scenarios [3]. This is consistent with findings in the radar-based activity-sensing literature, where micro-Doppler CNN approaches similarly trade off accuracy against deployment complexity [4], [5].

Traditional ultrasonic detection approaches commonly rely on signal processing operations and manually selected thresholds or statistical characteristics. These approaches can be effective when the sensing environment is controlled, but their performance can degrade when the relationship between the measured signal and the target changes. This motivates the use of data-driven classification methods that can learn discriminative patterns directly from ultrasonic measurements.

B. Machine Learning for Enhanced Classification

Machine learning provides an alternative to manually designed decision rules by learning relationships between measured ultrasonic signals and their associated classes. Previous work has demonstrated increasing interest in combining ultrasonic measurements with supervised machine learning methods, including feature extraction, feature selection, classifier optimization, and deep learning. Such approaches can improve the ability to associate ultrasonic measurements with relevant target or process information.

For ultrasonic data, classical machine learning approaches generally require the transformation of the raw measurement into a set of descriptive features. Examples include amplitude statistics, frequency domain characteristics, energy measures, and temporal descriptors. These features provide a compact representation that can subsequently be classified using methods such as linear models, Random Forests, or gradient boosting algorithms.

A limitation of this approach is that the selected features determine which information is available to the classifier. Features containing absolute amplitude or timing information may provide strong predictive performance under a particular recording configuration while also encoding properties associated with sensor-target distance or environmental geometry. Conversely, normalization can reduce the influence of such factors but may also remove information that is useful for classification.

Deep learning provides a complementary approach in which feature representations can be learned directly from structured signal inputs. CNN models [6] have been widely investigated for signal and ultrasound related classification problems, with the broader literature showing the increasing use of machine learning for ultrasound detection and classification tasks [7].

In the present study, both approaches are investigated. Classical models operate on engineered ultrasonic features, while CNN2D and CNN1D models learn from higher dimensional signal representations and waveform data. This makes it possible to distinguish the effect of the classifier from the effect of the underlying signal representation. This representation-dependence is not unique to ultrasonic sensing in the audio classification literature, raw waveform CNN architectures and spectrogram-based CNN architectures have been directly compared and shown to encode substantially different information, with neither representation uniformly dominant across tasks [8]–[10] a pattern consistent with the CNN1D/CNN2D divergence on the ChairTable condition reported in Section V-C. Prior work combining ultrasonic sensing with machine learning for related occupancy tasks, such as entrance counting, has likewise shown that classifier choice alone does not determine performance, reinforcing the need to evaluate representation and architecture as separate factors [11].

C. Red Pitaya and SRF02 for Ultrasonic Acquisition

The Red Pitaya STEMlab platform is a compact, software defined measurement and signal processing board suited to

high speed acquisition and generation of electrical signals. The STEMlab 125-14 variant, for example, provides two 14-bit analog inputs and two 14-bit analog outputs at up to 125 MS/s, with analog bandwidth extending to 60 MHz, combining an ARM processor with FPGA acquisition logic and supporting control through Python, MATLAB, and LabVIEW. The STEMlab 125-14's suitability as a low cost, FPGA backed acquisition front end for high speed sensor signals has been demonstrated directly in prior work combining it with an ARM+FPGA processing chain for sensor data acquisition and signal processing [12].

In the system used for this study, Red Pitaya's role is specifically *acquisition*, not signal generation. The ultrasonic ping itself is produced by the SRF02 ranging sensor's own transducer driver circuitry, triggered over an I^2C command issued from the Red Pitaya host; Red Pitaya's analog outputs are not used. Following each trigger, Red Pitaya performs triggered, DMA-based acquisition of a single input channel via its AXI mapped acquisition buffer, capturing the raw reflected echo at an effective sampling rate of approximately 1.953 MS/s (a decimation factor of 64 applied to the board's 125 MHz master clock) with 14-bit resolution. This acquisition path SRF02-generated excitation with Red Pitaya digitized echo is distinct from configurations in the literature where Red Pitaya's own analog output generates the acoustic excitation signal; that capability exists on the platform but is not exercised in this work.

Rather than treating the acquisition hardware as the primary contribution, this study focuses on how the acquired ultrasonic waveform can be represented and classified under environmental and positional variation. This distinction matters because, as the results in Section V and beyond demonstrate, the same raw acquisition can yield substantially different classification outcomes depending entirely on the downstream signal representation and preprocessing strategy not on the acquisition hardware itself.

D. Research Gap

Although ultrasonic sensing and machine learning have both been extensively studied, several questions remain open for human detection under changing environmental and positional conditions.

First, many machine learning evaluations focus primarily on overall classification accuracy. An aggregate figure can conceal substantial condition specific failures when a dataset mixes conditions of very different difficulty. A model may therefore appear highly accurate overall while failing systematically on one particular non human object or environmental configuration a pattern this study identifies directly in the ChairTable condition.

Second, the effect of signal representation is not equivalent to the effect of classifier architecture. Classical engineered features, spectrogram like representations, and raw waveforms expose fundamentally different information to a learning algorithm. Comparing only "classical machine learning" against "deep learning" therefore does not fully explain *why* one system generalizes better than another—the representation itself must be varied independently of the model family to isolate the cause.

Third, absolute amplitude and positional information can introduce spurious correlations between the measured signal and the experimental configuration rather than the target of interest. Removing or normalizing this information may improve environmental robustness, but can simultaneously discard information genuinely useful for recognizing a particular human activity. This trade off requires explicit, empirical evaluation rather than an assumption that normalization is uniformly beneficial.

Fourth, non human conditions are not necessarily equally difficult. An empty room reflection and a nearby object such as a chair or table can produce substantially different ultrasonic responses—plausibly because a chair's irregular, near-field reflection more closely resembles a human body reflection than a distant, flat wall does. Treating all non human measurements as a single homogeneous class, as much prior work implicitly does, risks hiding exactly this kind of condition-specific failure mode.

Finally, the distinction between human activities also requires direct investigation. If Standing and Reading can be separated primarily from positional information (for instance, posture height alone), a model may achieve high apparent accuracy without learning any activity specific signal characteristic at all. Comparing position aware and position-normalized representations provides a direct way to test this possibility, rather than assuming it away.

The research gap addressed by this work is therefore the absence of a systematic comparison of representation, normalization, architecture, and environmental generalization within a single, common ultrasonic human detection framework. Rather than evaluating one classifier under one controlled condition, this study evaluates multiple representations and model families under conditions specifically designed to expose environmental and positional dependencies that a narrower evaluation would miss. More broadly, this gap mirrors a recognized open problem in sensor based human activity recognition as a whole, where recent survey work identifies environment, subject, and sensor position shifts as a primary barrier to real world deployment across modalities, not only ultrasonic sensing [13], [14].

E. Research Contributions

The main contributions of this work are as follows:

1) A Systematic Ultrasonic Human Detection Framework: A complete sensing and classification pipeline is developed using a Red Pitaya measurement platform for ultrasonic acquisition system. The framework enables raw ultrasonic measurements to be processed using both classical machine learning and deep learning approaches.

2) Comparison of Multiple Signal Representations: The study compares a full feature representation, shape invariant engineered features, position aware CNN representations, position normalized CNN representations, and raw waveform

CNN processing. This comparison enables the effects of signal representation to be examined separately from those of the classification architecture.

3) Evaluation of Environmental Generalization: The experiments explicitly evaluate generalization between different environments rather than relying exclusively on randomly mixed training and testing observations. This evaluation exposes environment dependent failures that may remain hidden when training and testing data have similar environmental characteristics.

4) Condition Level Analysis of Human and Non Human Targets: Performance is reported separately for the *Empty-Room*, *ChairTable*, *HumanReading*, and *HumanStanding* conditions. This analysis demonstrates that non human conditions are not equally difficult and identifies *ChairTable* as a particularly challenging condition for environmental generalization.

5) Investigation of Positional Normalization: Position aware and position normalized representations are compared to investigate how reducing positional information affects both environmental robustness and human activity classification.

6) Analysis of the Standing vs Reading Problem: The study evaluates whether *Standing* and *Reading* can be separated primarily through positional information. The experimental results show that the position-aware representation does not provide a uniformly dominant solution, motivating further investigation into activity specific waveform information.

7) Identification of Representation Dependent Generalization Behavior: The experiments demonstrate a substantial difference between CNN1D and CNN2D performance on the challenging *ChairTable* condition. This indicates that the choice of signal representation can have a substantial effect on generalization, in addition to the choice of learning architecture.

8) Identification of Normalization Trade Offs: The results reveal that representations designed to reduce positional or amplitude dependence can improve performance for difficult environmental conditions while simultaneously reducing *HumanReading* classification accuracy. This trade off provides an important direction for future work on robust ultrasonic human detection.

III. Methodology

This section describes the complete methodology used for ultrasonic human detection and activity classification. The proposed framework consists of six major stages, shown in Fig. 1: ultrasonic data acquisition, preprocessing, feature representation, model training, model selection, and independent testing. The overall objective is to investigate how signal representation, positional normalization, and model architecture affect human detection and environmental generalization.

A. Overall Experimental Framework

The experimental framework is designed to evaluate ultrasonic sensing under different environmental and human-activity conditions. The complete processing pipeline is illustrated conceptually as

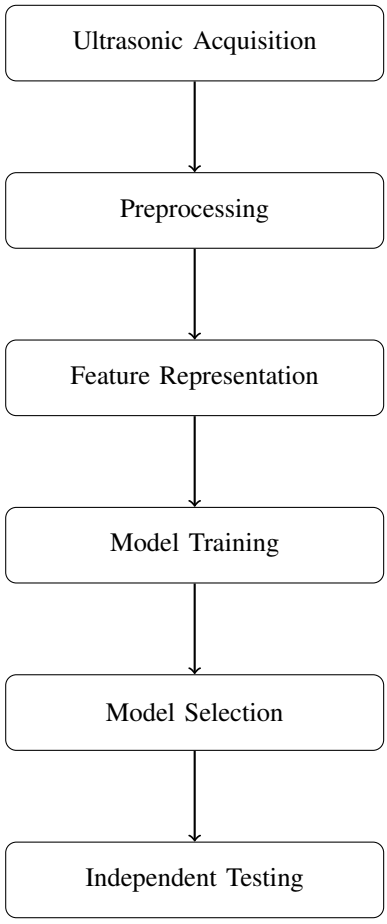


Fig. 1. Workflow of the ultrasonic based machine learning approach.

The framework incorporates both classical machine learning (ML) models and convolutional neural networks (CNNs). Classical ML models operate on engineered ultrasonic features, whereas the CNN models operate on structured signal representations and waveform data.

The experiments are organized around multiple representations in order to distinguish the effect of the classifier from the information available in the input representation. In particular, the study considers a full representation, shape invariant engineered features, position-aware CNN representations, position-normalized CNN representations, and raw-waveform CNN processing.

B. Ultrasonic Sensing and Data Acquisition

The ultrasonic sensing system uses a Red Pitaya for acquisition platform to generate and acquire ultrasonic signals. The transmitted ultrasonic signal propagates through the environment and is reflected by objects and human targets within the sensing region. The received signal therefore contains information related to the reflecting surface, target geometry, distance, and motion.

For each measurement, the received ultrasonic waveform is recorded and stored for subsequent processing. The acquisition

process is kept consistent across the experimental sessions to reduce variations caused by the measurement procedure.

The recorded waveform can be represented as

$$x[n], \qquad n = 0, 1, \ldots, N-1, \tag{1}$$

where $x[n]$ denotes the sampled ultrasonic signal and N represents the number of samples in an individual measurement window.

The recorded signal is subsequently processed into representations appropriate for the different classification approaches.

C. Experimental Conditions

The dataset contains both human and non human conditions. The primary conditions considered in the human detection experiments are:

- *EmptyRoom*: an environment without a human target.
- *ChairTable*: a non human condition containing the relevant static furniture configuration.
- *HumanReading*: a human subject performing a reading activity.
- *HumanStanding*: a human subject in a standing configuration.

These conditions are intentionally analyzed separately rather than being combined into a single "non-human" class. In particular, *EmptyRoom* and *ChairTable* produce substantially different classification behavior and therefore provide different tests of environmental robustness.

The data set also contains measurements from multiple human subjects. To protect subject privacy, individuals are identified throughout this paper only by anonymized labels (Person 1, Person 2, Person 3, Person 4, Person 5) rather than by name; the subject-level analysis in Section V-E involves Person 3 and Person 4, the two subjects recorded in Env2 as part of the main evaluation pipeline. The subject-level analysis is performed separately to investigate whether the observed classification behavior is consistent between subjects.

D. Onset Detection and Alignment

Exploratory analysis of the raw waveforms (Section V-B) confirms that reflected energy in every condition is concentrated in a short echo burst against an otherwise near-silent background, rather than being spread evenly across the measurement window. This structural property motivates an onset-detection and alignment procedure, used to construct the position-normalized representations described in Section IV-D.

For each recording, the burst onset is located as the first sustained crossing of a robust, per recording noise floor threshold applied to the Hilbert envelope of the signal, rather than the global peak of the waveform. The threshold is set at the envelope's median plus k times its median absolute deviation (MAD), and a minimum duration requirement is applied so that brief noise spikes are not mistaken for the echo onset. This first-arrival approach was adopted in preference to a simple global-amplitude-peak detector because the strongest reflection in a window is not always the nearest one: a more distant, acoustically harder surface can reflect more strongly than a closer, softer target, which would bias a peak-based onset estimate toward the wrong reflector. Once located, each recording is cropped to a fixed-length window anchored at its own detected onset, forming the aligned input used by the position-normalized representations.

E. Hardware and Software Components

The Red Pitaya STEMlab platform (Fig. 2), which incorporates a 14-bit, 125 MS/s ADC appropriate for recording high-resolution ultrasonic Doppler echoes, serves as the foundation for the sensing front-end. An SRF02 ultrasonic transducer module, which is already a component of the sensing setup, is used in the ultrasonic interface. With support for both I^2C and UART communication interfaces, this sensor allows for flexible integration and operates at about 40 kHz. The sensor is appropriate for short-range indoor occupancy monitoring because of its effective measurement range, which is roughly 15 cm to 6 m. While the paired receiver records the reflected echo signal, one transducer element acts as the transmitter. The Red Pitaya then digitizes the signal for additional processing.

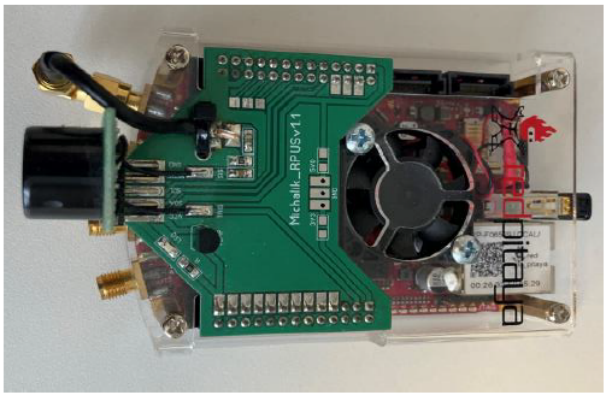


Fig. 2. Red Pitaya STEMlab platform used for ultrasonic signal acquisition and streaming.

All data processing, model training, and evaluation were implemented in Python 3.13.7. The pipeline follows a standard scientific-computing and machine learning stack.

NumPy (2.5.3) and Pandas (3.0.6) were used throughout for numerical array operations and tabular dataset management respectively, and feature table construction. SciPy (1.18.1), specifically its signal module's short time Fourier transform (scipy.signal.stft), was used to compute band limited spectrograms as input to the two dimensional convolutional model.

Three classical machine learning classifiers were trained and evaluated using Scikit-learn [15] (1.9.1) i.e a linear model (logistic regression), a Random Forest ensemble [16], and a Gradient Boosted Decision Tree model implemented via the XGBoost library [17] (3.4.1). Model performance was quantified using Scikit-learn's metrics module (accuracy, precision, recall, F1-score, and ROC-AUC).

Deep learning models were implemented in PyTorch [18] (2.14.0), comprising two convolutional architectures—a one

dimensional CNN (CNN1D) operating directly on raw time domain ultrasonic waveforms, and a two dimensional CNN (CNN2D) operating on the corresponding band limited spectrogram representation. Training was performed using the Adam optimizer [19] with binary cross entropy loss and early stopping based on validation loss, executed on an Apple MacBook Pro (M3 chip) using PyTorch's Metal Performance Shaders (MPS) backend for GPU accelerated training.

To prevent data leakage, all models (classical and deep learning alike) were trained and hyperparameter tuned exclusively on the Env1 train/validation split, after which the trained parameters were frozen and serialized to disk (.joblib for classical models, PyTorch .pt checkpoints for CNNs). Final performance was then evaluated once on the independent Env2 test set by loading these frozen models, ensuring the test data played no role in model fitting or selection.

F. Full Feature Representation

The full representation retains the available signal derived information without applying the shape invariant normalization used in the corresponding classical experiment. This representation therefore preserves absolute signal characteristics that may contain information related to amplitude, timing, and measurement geometry.

The full representation is used as a baseline against which normalized representations can be compared.

Although retaining additional information can improve classification within a particular environment, it can also allow a model to exploit correlations between signal characteristics and the experimental configuration. Therefore, the full representation provides an important baseline for evaluating environmental generalization.

G. Classical Machine Learning Models

Several classical supervised learning algorithms are evaluated using the engineered feature representations. The evaluated models include a linear classifier, Random Forest (RF), and Extreme Gradient Boosting (XGBoost).

The linear model provides a relatively simple baseline, while Random Forest and XGBoost provide nonlinear decision boundaries and can model interactions between engineered features.

For a feature vector $\mathbf{f}$ and class label y, the classifier learns a mapping

$$\hat{y} = g(\mathbf{f}; \boldsymbol{\theta}), \tag{2}$$

where $\boldsymbol{\theta}$ represents the parameters learned during training.

Model selection is performed using the validation partition. The test partition is then evaluated using the selected model without further parameter fitting.

Because multiple classical models are evaluated, the condition level results reported for the classical shape invariant representation use the best performing classical model for the corresponding comparison. This avoids selecting an arbitrary model based on file ordering or model name ordering.

H. CNN Representations

In addition to engineered classical features, convolutional neural networks are evaluated to determine whether learned representations can capture information that is not explicitly encoded by the engineered features.

Two CNN architectures are considered:

- CNN1D, which processes the ultrasonic waveform along the temporal dimension.
- CNN2D, which processes a two dimensional representation of the ultrasonic signal.

Both architectures follow a conventional convolutional block structure of convolution, batch normalization [20], ReLU activation, and pooling, repeated across several layers before a fully connected classification head that includes a dropout layer for regularization batch normalization in particular was used to stabilize training given the relatively small per condition batch sizes available after the Env1/Env2 split.

The CNN models are evaluated using both position aware and position normalized representations where applicable.

IV. Implementation

A. Measurement Setup

The experimental measurement setup was established in a controlled indoor laboratory environment for the acquisition of ultrasonic sensor measurements. The ultrasonic sensor was mounted on a tripod at a fixed height of approximately $2.1\ \mathrm{m}$ above the ground and positioned toward the measurement area. The fixed mounting position and orientation provided consistent measurement geometry throughout the experiments and enabled the sensor to capture reflected ultrasonic signals from objects within the monitored area.

The ultrasonic transducer was connected to the Red Pitaya STEMlab board, which was used as the data acquisition unit. The analog input channels of the Red Pitaya were used to acquire the sensor signals. The Red Pitaya was connected to a host laptop via Ethernet, providing the communication interface required for data transfer and SSH tunneling with remote access and control of the measurement hardware.

The acquired sensor data was transferred to the host computer for storage and subsequent processing. During data collection, the sensor position, mounting height, and orientation were kept fixed to minimize variations in the measurement conditions. The surrounding indoor environment, including the corridor walls, doors, tiled floor, and other stationary structures, remained part of the measurement environment and therefore contributed to the recorded ultrasonic reflections.

Figure 3 shows the complete measurement arrangement, including the tripod mounted ultrasonic sensor, the Red Pitaya STEMlab data acquisition hardware, the host laptop, and the associated power and communication connections.

B. System Configuration

A C process with UDP server program on the RedPitaya (`dma_with_udp_faster`) continuously sends analog to digital converted (ADC) ultrasonic waveforms to the host



Fig. 3. Experimental ultrasonic sensing setup in the indoor corridor environment. The Red Pitaya based sensing unit is mounted on a tripod and positioned along the corridor, while a laptop is used for signal acquisition, monitoring, and data recording.

computer. High speed UDP sockets are used for data transfer, and SSH is used for control commands. The following was how the system parameters were set up:

- ADC Master Clock Rate: 125 MS/s; effective ultrasonic sampling rate ≈1.953 MS/s after the decimation factor of 64 described in Section II-C
- Samples per Frame: 25,000
- Transmission Protocol: UDP over 169.254.x.x local network
- Data Logging Format: Timestamped CSV files per sample
- CNN Architectures: 1D CNN (raw waveform, 25,000 or 4,900 samples) and 2D CNN (band limited spectrogram, 5 frequency bins $\times$ variable time bins)
- Model Serialization Format: `.joblib` (classical models), PyTorch `.pt` checkpoints (CNNs)

All data processing, model training, and evaluation were performed on an Apple MacBook Pro (M3 Pro, 18-GB unified memory). The convolutional neural network models (CNN1D, CNN2D) were trained using PyTorch's Metal Performance Shaders (MPS) backend for GPU accelerated training. The classical machine learning models (linear, Random Forest, XGBoost) were trained on CPU via scikit-learn and XGBoost, as neither library provides GPU or MPS support on Apple Silicon.



(a) Environment 1 (laboratory room)



(b) Environment 2 (corridor)

Fig. 4. Measurement setup in Environment 1 and Environment 2, illustrating the physically distinct spaces used to evaluate cross-environment generalization.

C. Data Collection and Flow

The data collection process was designed to evaluate the proposed ultrasonic sensing system's ability to distinguish human presence and activity under varying environmental conditions. Ultrasonic measurements were acquired using the Red Pitaya based sensing platform described in Section III and recorded for subsequent offline processing. Collected data were organized by experimental condition, subject, and dataset split to maintain a consistent evaluation procedure across all experiments.

Four experimental conditions were recorded: *EmptyRoom*, *ChairTable*, *HumanReading*, and *HumanStanding*. *EmptyRoom* represents the absence of a person and establishes a baseline environmental background response. *ChairTable* represents a static, non-human scene containing furniture, used to evaluate whether the system can distinguish stationary environmental objects from human presence. *HumanReading* and *HumanStanding* represent two distinct human postures and activity patterns, enabling evaluation of the system under different human behaviors; Fig. 5 shows the *HumanReading* condition as recorded.



Fig. 5. *HumanReading* condition in Env1: a seated subject reading, directly beneath the tripod mounted sensor rig. The laptop screen shows the live acquisition tool with the captured ultrasonic waveform.

Figure 6 summarizes the resulting data flow. Raw ultrasonic waveforms acquired by the sensing platform are transferred to the host computer and stored as uncompressed signal samples. These are subsequently converted into the representation required by each classification approach engineered features or shape-invariant features for the classical models, and raw one-dimensional waveforms or two dimensional signal representations for CNN1D and CNN2D respectively. Position-aware and position-normalized variants are generated where applicable (Section IV-D).

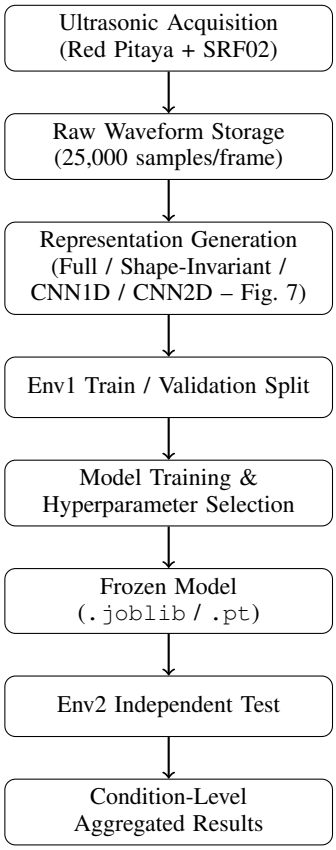


Fig. 6. End to end data flow from ultrasonic acquisition through representation generation, Env1 only training and freezing, and independent Env2 testing. All four representations are generated from the same raw waveform storage and follow an identical train/freeze/test procedure.

Each representation is then partitioned into the predefined training, validation, and test subsets established during data collection, with separate processing pipelines maintained for the classical models, CNN1D, and CNN2D to accommodate their differing input formats. Frozen models are evaluated once against the independent test set, and predictions are aggregated into the condition-level and environment-level results reported in Section V.

This data-flow architecture ensures that every classification approach is evaluated under an identical acquisition and testing procedure, isolating the effects of signal representation, normalization, environmental condition, and human activity as independent experimental variables.

D. Data Pre-Processing and Model Training

1) Data Pre-Processing: Each recorded sample was associated with its experimental condition and dataset split prior to model training. Preprocessing was designed to produce a consistent input format across experiments while preserving the signal characteristics relevant to each representation.

For the classical machine learning models, ultrasonic signals were converted into two engineered feature sets a full representation retaining absolute signal statistics, and a shape invariant representation computed from relative statistical and spectral characteristics: standard deviation relative to peak amplitude, peak to peak amplitude relative to peak, crest factor, dominant frequency, and band energy fraction.

For the deep learning models, ultrasonic data were retained as structured signal inputs rather than reduced to scalar features. CNN1D receives the raw waveform as a one dimensional input, allowing the network to learn temporal patterns directly from the signal. CNN2D receives a two dimensional spectrogram representation, allowing spatial and structural patterns across time and frequency to be learned jointly.

Where applicable, each representation was generated in both a position aware and a position normalized form. The position aware form retains the signal's absolute positional and amplitude information as measured the position normalized form reduces this influence through amplitude and timing relative normalization. The two forms were evaluated separately (Section V) to isolate the effect of positional information on classification performance.

All preprocessing operations were applied identically to the training, validation, and test partitions. Test data remained isolated from every stage of model development.

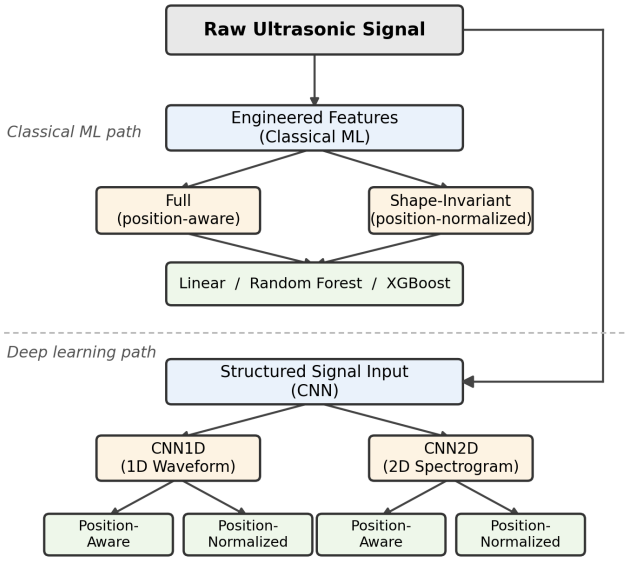


Fig. 7. Representation taxonomy: each input representation (full, shape-invariant, CNN1D, CNN2D) is paired with its corresponding model family...

2) Model Training: Three classical classifiers Linear, Random Forest, and XGBoost were trained on the engineered feature representations, providing a linear baseline and two nonlinear ensemble approaches for comparison. Two convolutional architectures, CNN1D and CNN2D, were trained on the corresponding waveform and spectrogram representations; where both position aware and position normalized inputs were available, a separate model was trained for each.

Each model was trained on the designated training partition, with hyperparameter selection and early stopping (for the CNN architectures) performed using the validation partition. The selected configuration for each model and representation was then frozen and serialized to disk, and evaluated exactly once against the independent test partition.

Applying this identical training and freezing procedure across every classifier and representation ensures that differences in reported performance can be attributed to the choice of model and signal representation, rather than to inconsistencies in the underlying training procedure.

V. Results and Discussion

A. Dataset Summary

The final dataset comprises 70,000 recordings: 30,000 in Env1 and 40,000 in Env2 (30,000 main pipeline recordings plus 10,000 held-out recordings from a third Env2 subject, Person 5, excluded from the main evaluation to preserve a symmetric two-subject design across both environments). Across the main pipeline, 50,000 recordings are labeled *HUMAN* and 20,000 *NON_HUMAN*.

B. Exploratory Signal Analysis

Representative raw waveforms, FFT magnitude spectra, and spectrograms for each condition (Fig. 8) confirm a consistent ≈40 kHz spectral peak across all four conditions, together with a pulse echo structure in which reflected energy is concentrated in a short burst against an otherwise near silent background the structural property motivating the onset-detection and alignment procedure in Section III-D.

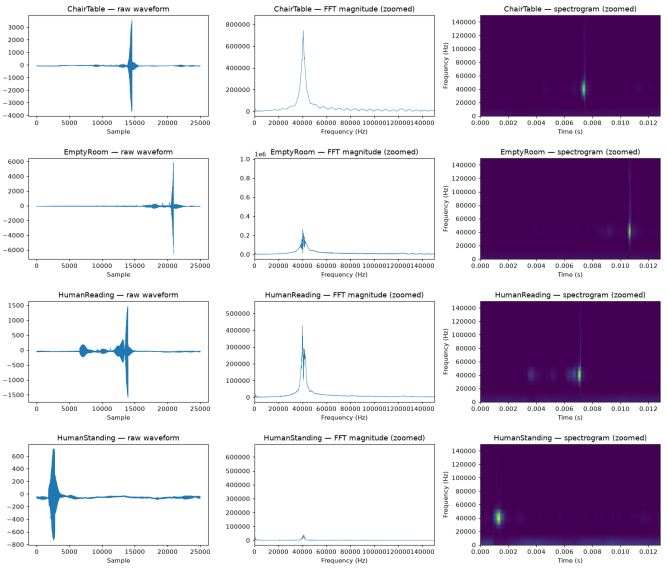


Fig. 8. Representative raw waveforms, FFT magnitude spectra, and spectrograms for each condition. A consistent ≈40 kHz spectral peak is visible across all four conditions, with reflected energy concentrated in a short pulse echo burst against a near silent background.

TABLE I
HUMAN-PRESENCE DETECTION: ENV2 TEST ACCURACY (%), NON-HUMAN CONDITIONS

Representation	ChairTable	EmptyRoom
Full	0.00	96.42
Aligned	69.91	95.69
Shape	55.14	97.21
2D-A	22.48	96.12
2D-N	33.53	93.65
1D-A	93.02	**98.00**
1D-N	**97.72**	93.71

Full = hand-crafted features, position aware; Aligned = hand-crafted features on the onset-aligned waveform, without the amplitude normalization applied in Shape; Shape = shape invariant features (timing and amplitude normalized); 2D-A/2D-N = CNN2D position-aware/normalized; 1D-A/1D-N = CNN1D position-aware/normalized. For Full, Aligned, and Shape, the value is the best of the three classical models per condition. Bold = best per condition.

A further exploratory finding (Fig. 9) is that raw signal RMS is *higher* for the non human conditions than for either human condition (EmptyRoom highest, followed by ChairTable, then HumanReading, then HumanStanding), the opposite of a naive expectation that human presence increases reflected energy. A plausible explanation is that hard, flat, distant surfaces (an empty wall) and rigid nearby objects (a chair) produce a stronger, more coherent specular reflection than a human body's soft, irregular, scattering surface.

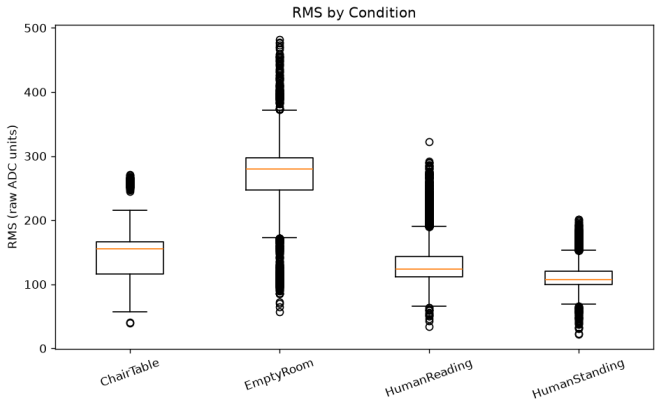


Fig. 9. Raw signal RMS by condition, ordered from highest to lowest: EmptyRoom, ChairTable, HumanReading, HumanStanding.

C. Human Presence Detection and Representation Comparison

Tables I and II report Env2 test accuracy for each condition, across all seven representation/branch combinations (best classical model shown where multiple classical models were evaluated under one representation).

Three of the four conditions generalize well (roughly 86 to 100%) under nearly every representation. *ChairTable* (Fig. 10) is the exception: under the full, position aware representation, all three classical models misclassify *every* ChairTable recording as *HUMAN* (0.00% accuracy), despite 99.93% validation accuracy on Env1 the characteristic signature of a model exploiting an environment specific shortcut rather than a transferable signal. Fig. 11 summarizes this condition comparison across every representation tested.

Removing absolute amplitude information (shape invariant features), absolute timing information alone (the aligned representation), or both via CNN position normalization each substantially improves ChairTable accuracy relative to the position aware baseline (Fig. 12). The improvement is not uniform across representations: CNN1D reaches 93.02% even in its position aware form and 97.72% once position normalized,

TABLE II
HUMAN-PRESENCE DETECTION: ENV2 TEST ACCURACY (%), HUMAN CONDITIONS

Representation	HumanReading	HumanStanding
Full	99.51	99.97
Aligned	92.73	97.34
Shape	90.89	94.08
2D-A	**99.93**	**99.98**
2D-N	96.36	99.74
1D-A	98.59	99.97
1D-N	86.06	99.91

Representation abbreviations as in Table I. Bold = best per condition.



Fig. 10. The ChairTable condition: an empty chair placed beneath the sensor, the non-human condition responsible for the representation-dependent generalization failure discussed in this section.

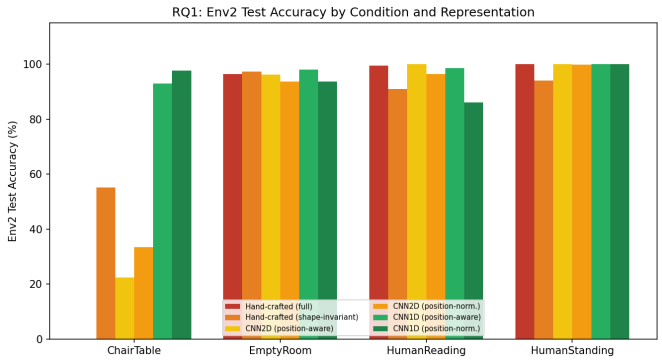


Fig. 11. Condition comparison across representations.

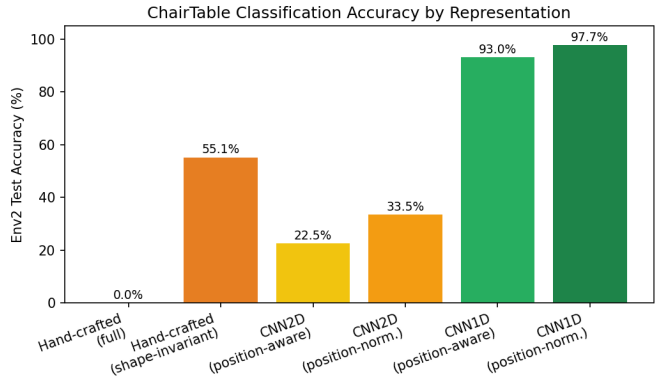


Fig. 12. ChairTable accuracy progression across representations and models.

while CNN2D improves only to 22.48%/33.53% under the same two branches. A plausible explanation is that CNN2D's band limiting to 20 to 60 kHz, while justified for isolating the carrier, discards broadband transient shape information that CNN1D retains and that may specifically distinguish a chair's reflection from a human body's.

The overall confusion pattern for the strongest configuration (CNN1D, position-normalized) is shown in Fig. 13. A second pattern is that *HumanReading* accuracy declines under every position/amplitude-normalized representation relative to its position-aware counterpart (shape-invariant: 90.89% vs. full's 99.51%; CNN1D normalized: 86.06% vs. CNN1D-aware's 98.59%; CNN2D-normalized: 96.36% vs. CNN2D-aware's 99.93%), while *HumanStanding* and *EmptyRoom* remain stable. Recurring across three independent model families, this is unlikely to be noise plausibly, part of what made Reading recordings easy to recognize as *HUMAN* under unnormalized representations was a positional artifact of where a seated subject happened to sit, rather than the page-turning motion signature itself.

A third pattern emerges from the *aligned* representation, which isolates the effect of timing normalization alone, independent of the amplitude normalization also applied in *shape-invariant*. Under *aligned*, the three classical models diverge sharply in robustness: Random Forest and XGBoost reach

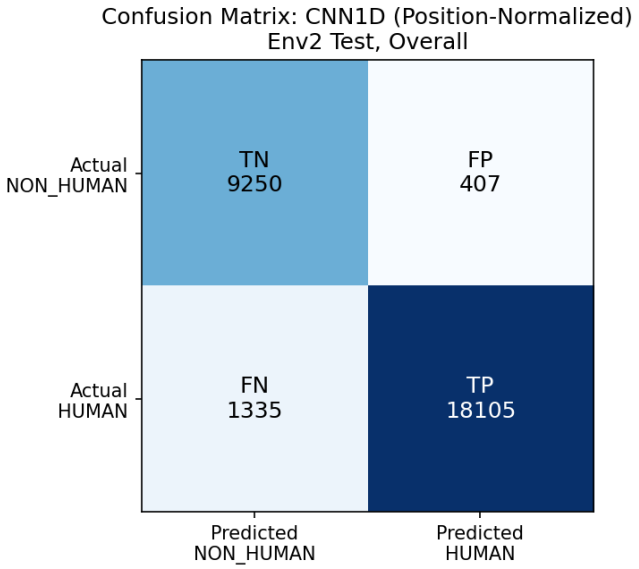


Fig. 13. Confusion matrix for CNN1D (position-normalized), Env2 test set, overall.

TABLE III
STANDING VERSUS READING: ENV2 TEST ACCURACY (%) BY MODEL AND BRANCH

Model	Position-Aware	Position-Normalized
Linear	81.68	79.57
Random Forest	87.43	88.95
XGBoost	88.73	88.76
CNN1D	91.58	**95.30**
CNN2D	**97.13**	96.86

99.31% and 99.38% Env1 validation accuracy respectively, yet their Env2 ChairTable accuracy collapses to 20.31% and 20.83% while the simpler linear model, despite *lower* validation accuracy (95.45%), generalizes far better on the same representation (69.91% on ChairTable, exceeding even the best shape invariant result of 55.14%). This is the same near perfect validation/collapsed generalization signature observed for the *full* representation above, now recurring *within a single, timing-normalized representation*, split specifically by model flexibility. It suggests model capacity itself, not only signal representation, affects susceptibility to environment-specific shortcuts: Random Forest and XGBoost's greater capacity to fit nonlinear feature interactions appears to make them more, not less, prone to exploiting a residual positional or amplitude cue once timing alone is normalized, whereas a simpler linear model is forced toward a more genuinely transferable decision boundary.

D. Standing versus Reading Classification

Table III reports Env2 test accuracy for the Standing versus Reading task, by model and branch.

Fig. 14 plots the same accuracy figures by model and branch. This task was expected to expose a trivial shortcut

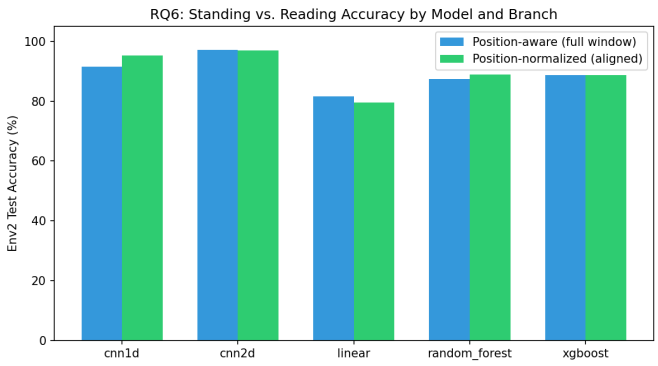


Fig. 14. Standing versus Reading Env2 test accuracy by model and branch.

under the position aware branch, since Standing and Reading differ substantially in posture height. Real data results do not bear this out classical models and CNN2D show near identical performance between branches, and CNN1D's position *normalized* result (95.30%) exceeds its position aware result (91.58%) the opposite of what a dominant posture height shortcut would predict. Two explanations are plausible and not mutually exclusive: posture height may carry less separating power for this distinction than assumed; and, for CNN1D specifically, the shorter, precropped 4,900-sample aligned window may simply be an easier input to learn from than the mostly-silent 25,000-sample full window, independent of confound removal.

E. Subject Consistency

For the Standing versus Reading task, Person 3 is classified more accurately than Person 4 under *every* representation tested (e.g., CNN2D position aware: 99.33% vs. 94.93%; shape invariant Random Forest: 93.10% vs. 84.57%), indicating a genuine subject-level effect rather than noise.

For human presence detection, the pattern is more structured: Person 3 outperforms Person 4 under both classical representations (full: 100.0% vs. 98.41–99.48%; shape-invariant: 94.06–95.08% vs. 87.81–89.82%), while Person 4 outperforms Person 3 under all four CNN representations, though CNN2D margins are negligible ($<$0.02 in both branches). This suggests the subject level factor driving the consistent Person-3 advantage on Standing versus Reading is specific to that distinction rather than a general "Person 4 is harder to recognize as human" effect.

F. Discussion

Three findings form the central contribution of this work. First, non human conditions are not interchangeable *Empty-Room* generalizes reliably under every representation, while *ChairTable* exposes a severe, representation dependent failure ranging from complete misclassification to near perfect accuracy depending entirely on signal representation invisible under a single aggregate *NON_HUMAN* accuracy figure. Second, representation affects generalization as much as, or more than, the classical versus deep learning distinction CNN2D, despite being a deep architecture, generalizes substantially worse than CNN1D on the hardest condition. Third, normalization is not uniformly beneficial every representation that improves ChairTable robustness simultaneously reduces HumanReading accuracy, and the trivial baseline expectation for Standing versus Reading was directly contradicted by real data.

VI. Conclusion and Future Work

This work evaluated ultrasonic human presence and activity classification across seven representation/branch combinations and five classifiers (linear, Random Forest, XGBoost, CNN1D, and CNN2D) under an explicit cross environment generalization protocol. The central finding is that aggregate accuracy figures substantially overstate reliability: the *ChairTable* condition exposes a severe, representation-dependent generalization failure from 0.00% accuracy under raw hand crafted features to 97.72% for a position normalized 1D CNN invisible in any single overall accuracy number. CNN1D operating on position normalized raw waveforms was the strongest configuration overall, outperforming both classical feature engineering models and the band limited CNN2D on the hardest condition. For the Standing vs Reading task, the assumption that a position aware representation would trivially dominate via posture height was not supported, and normalization that improved robustness on the hardest human detection condition consistently reduced accuracy on the Reading activity across both tasks.

Future work should prioritize three directions suggested directly by these results. First, the waveform shape information that allows CNN1D to succeed where CNN2D does not on ChairTable should be investigated directly, for example through learned filter visualization. Second, the Reading accuracy cost incurred by every normalized representation warrants targeted investigation, potentially through representations that remove only timing or only amplitude information rather than both simultaneously. Third, the held-out Person 5 and additional environments beyond Env1/Env2 would allow the generalization claims made here to be tested against a larger and more diverse set of unseen conditions.

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